

Mode Model And Fit Parameters

The reconstruction uses a Hermite-Gaussian-inspired beam prior. The diagnostics visualize low-order HG building blocks, while the current fit uses a fundamental HG00-like envelope with curvature and global phase.

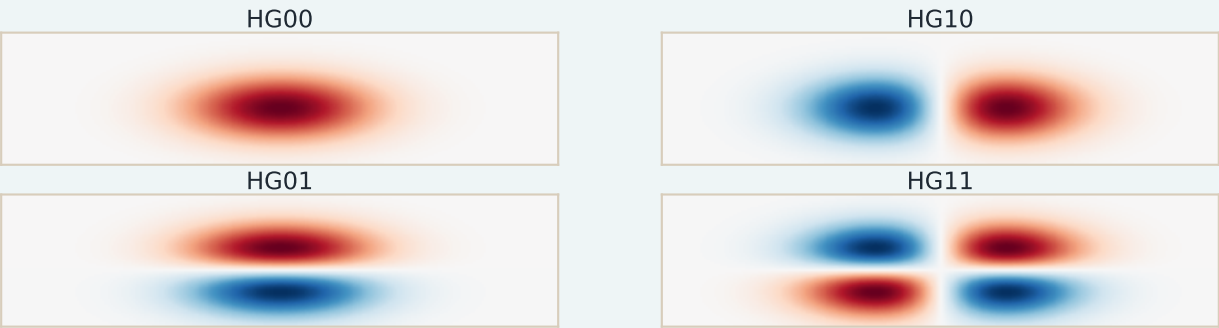
Model Form

$$E(x, y) = A \exp\left[-\frac{r^2}{s^2}\right] H(x, y) \exp(i[kr^2 + \phi_0])$$
$$r^2 = (x - x_0)^2 + (y - y_0)^2$$

Current fit: $H(x, y) = 1$, so only the HG00 envelope is used.

The reconstruction therefore solves for center, width, amplitude, curvature, and global phase.

Low-Order HG Building Blocks



Fit Parameters

x_0, y_0 : beam center.
 A and s : complex scale and transverse width.
 κ and ϕ_0 : phase curvature and global phase.
One parameter vector is fitted for each retained (z, ν) .

Live Fit Example At 19.5 GHz

$x_0 = -16.1$ mm, $y_0 = +13.3$ mm, $s = 49.268$ mm,
 $\kappa = -84.2$ 1/m², $\phi_0 = 1.69$ rad

Live HG00 Field From The Cached Fit

